

E-commerce Business Analytics

Executive Summary - Portfolio Project

Business Problem

Marketplace and e-commerce teams need to monitor revenue, customer demand, delivery quality, product performance, and review score in one integrated view. This project turns order-level marketplace data into KPI tracking, dashboard insights, and business recommendations.

Dataset & Transparency

This package uses synthetic Olist-style marketplace data with the same table structure as the public Olist Brazilian E-commerce dataset so the project can be run immediately. To use the real public dataset, replace the CSV files in data/raw with the Olist dataset from Kaggle and rerun the notebook/script.

Key Metrics

Metric	Value
Total Revenue	17,900,525.67
Delivered Orders	96,551
Unique Customers	96,551
Average Order Value	185.40
Average Review Score	3.83
Late Delivery Rate	24.18%
Average Delivery Days	9.15

Key Findings

- The highest-revenue category is computers_accessories, generating 1,685,925.43 in revenue.
- The top customer state is SP, generating 7,350,781.42 in revenue.
- Orders with lower review scores show higher late delivery rates, indicating that logistics performance is a major driver of customer satisfaction.
- Seller, category, and region performance should be monitored together because revenue growth can hide delivery or review quality issues.

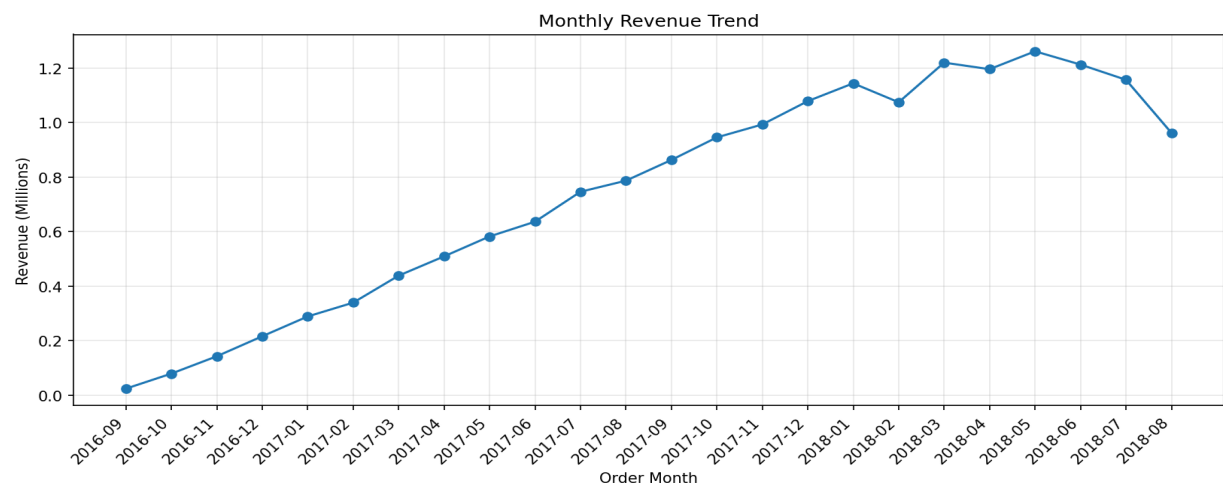
Business Recommendations

- Prioritize high-revenue categories for campaigns, stock planning, and seller quality monitoring.
- Monitor late delivery rate weekly, especially in high-revenue categories and states.
- Create seller performance tiers using revenue, review score, late delivery rate, and order volume.
- Focus marketing and logistics investment on high-demand states while testing growth in lower-penetration regions.
- Use the dashboard as a weekly business review tool for sales, operations, and customer experience teams.

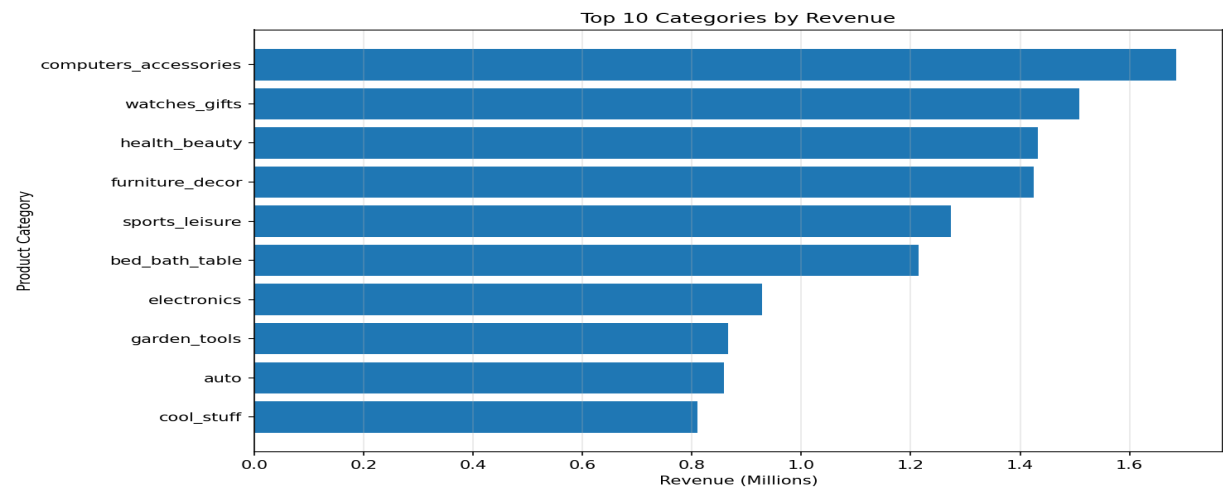
Visual Evidence

The following charts are generated from the analytical dataset and are included in the dashboard workbook.

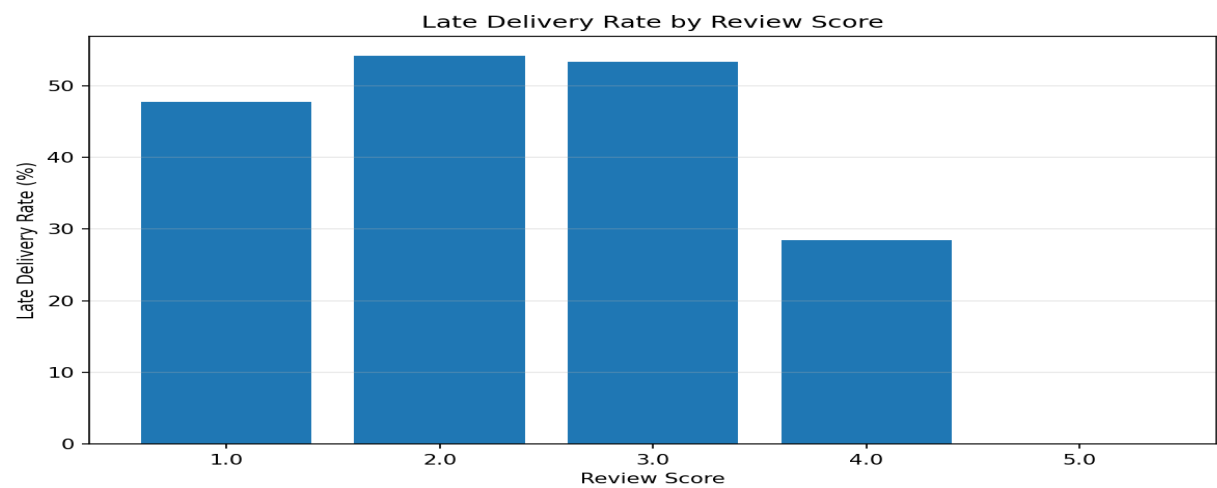
Monthly Revenue Trend



Top Categories by Revenue



Late Delivery Rate by Review Score



Portfolio Positioning

This project demonstrates practical Data Analyst skills: SQL KPI analysis, Python data cleaning and EDA, dashboard creation, business storytelling, and actionable recommendations. It is suitable for Data Analyst, Business Intelligence Analyst, Junior Data Analyst, Business Analyst, and Operations Analyst applications.

Suggested CV Bullet

Built an end-to-end e-commerce analytics project using SQL, Python, and Excel Dashboard to analyze 100K+ marketplace orders, identify revenue drivers, evaluate delivery performance, and develop business recommendations to improve customer satisfaction.